Product Journey Map · QuoteVoice

How a voice-to-quote AI product for painters went from a Reddit complaint to a live app, one stage at a time.

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Live: **saif-byte.github.io/quotevoice** · Code: **github.com/saif-byte/quotevoice**

STAGE	1 · Discover	2 · Validate	3 · Scope	4 · Prioritize	5 · Architect	6 · Database	7 · Quality	8 · Add features	9 · User test	10 · Ship
WHAT I DID the actual work	• Read Reddit + trade forums in two passes • Wrote each repeated pain as a problem • Scored a shortlist on 4 filters	• Mapped 9 assumptions by impact and evidence • Wrote interview script + voice test • Set pass / fail limits up front	• Defined one loop: record → read → check → price → send → win • Cut invoicing, CRM, photos, native app	• Ordered releases by risk • Core loop → trust → closing the job → safety	• Free stack, no servers to run • AI reads the note → code does the math • Fallback reader if AI is down	• 3 tables: profiles, quotes, AI usage • Row-level security everywhere • 3 narrow functions for homeowners	• Hand-labeled voice notes, scored field by field • 33 unit tests • In-app AI metrics vs targets	• Same 8 steps: need → map → data → build → test → secure → ship → verify • e.g. "tell me when a customer accepts"	• AI agent used the live app as painter "Mike" • Logged 15 issues • Fixed all before new features	• Push → test → build → deploy, automatically • Keep-alive + self-resetting demo • Weekly dependency checks
KEY DECISION the thinking	*"Pick a pain that's measurable and demos in 2 minutes."*	*"Will painters really talk a quote aloud? Test that first."*	*"The AI never sets a price."*	*"Ship the riskiest learning first, polish later."*	*"Nothing should break when the AI or the signal does."*	*"Rules live in the database, not the app."*	*"100% on tuned data means little. Hold some back."*	*"Data model first, then the smallest useful version."*	*"Fix trust before adding anything new."*	*"A failing test blocks the deploy."*
TOOLS touchpoints	R Reddit D Forums	AI PM agent D Project doc	D Build / skip list	D Roadmap	Re React TS TypeScript Mic Web Speech G Gemini S Supabase	PG Postgres S Auth + RLS	T Vitest ≡ Eval sets	S Realtime GH Git	AI Persona tester Live app	GH GitHub Actions GH Pages
OUTPUT what it produced	**Problem: quotes typed at night lose jobs. First market: US painters.**	**Assumption map, experiment E1, decision log, risk register.**	**Every feature mapped to a bet or a metric.**	**Release plan v1 → v1.6, each answering one question.**	**Architecture + typed data pipeline (RawJob → Quote).**	**Versioned migrations; quote stored as JSON + customer copy.**	**Three eval sets incl. a held-out one never tuned on.**	**Send & track, live notifications, onboarding, demo account.**	**v1.1: 15 fixes + new test cases from the misreads.**	**Live app, product video, demo account for anyone to try.**
PROOF one number	**4 filters** pain · measurable · demo · not crowded	**9 bets** ranked; riskiest tested first	**5 cuts** made on purpose, written down	**14** releases, each tested and deployed automatically	**$0** a month to run	**17 / 17** attack checks blocked	**97.4%** accuracy on unseen notes	**Hours** from request to live, per feature	**73 → 100%** accuracy on the notes it got wrong	**Minutes** from push to live

WHY THIS STACK

what it does, why I picked it

Chosen against four rules: **$0 to run** · **no servers to look after** · **keeps working if the AI or signal drops** · **any piece can be swapped**

Re React + TypeScript + Vite

The app painters and homeowners use, on any phone or laptop.

Why: Types catch quote and pricing bugs before users do; builds to static files that host free.

Picked over: plain JavaScript, a native app

Mic Web Speech API

Turns the painter's voice into text.

Why: Built into phone browsers: free, instant, nothing to install, no audio stored by the app.

Picked over: paid speech-to-text APIs

G Google Gemini

Reads the note into rooms, sizes and surfaces.

Why: Free tier plus strict JSON-schema output. Sits behind one interface, so the model can be swapped.

Picked over: paid LLM APIs; letting the AI price

S Supabase

Sign-in, database, live updates, server-side AI calls.

Why: Real Postgres with row-level security: data rules the app itself can't bypass. All in one free service.

Picked over: Firebase, a custom server

GH GitHub Pages + Actions

Hosting and the test-then-deploy pipeline.

Why: Free, and every push is tested before it goes live. Code, CI and hosting in one place.

Picked over: paid hosting, manual deploys

T Vitest

Unit tests and the AI accuracy eval.

Why: Same setup as the build tool, fast enough to run the whole eval on every push.

Picked over: manual spot checks

HOW IT'S KEPT SECURE

attack → protection

Rules live in the database, where a modified app can't skip them, and every one was attacked on the live system: **17 of 17 attack checks blocked.**

! Seeing another painter's quotes

Row-level security on every table: each account only ever reaches its own rows.

! Faking "quote accepted"

Only one database function can accept; it locks the row, refuses expired quotes, and a trigger stops painters editing it.

! Guessing customer links

16 random characters (about 92 bits) per link, format-checked; the link only reveals the customer copy.

! Stealing or draining the AI key

Key lives only on the server; sign-in required; 100 notes per painter and 1,500 per day in total.

! Injecting scripts (XSS)

React escapes all text, a Content Security Policy blocks outside scripts, and link data is cleaned before display.

! Clickjacking

The app refuses to run inside another site's frame.

! Flooding the free database

Size limits on every field, 5,000 quotes per painter, 4,000-character notes.

! Vulnerable packages, leaked secrets

0 known vulnerabilities, weekly Dependabot checks, no secrets in the code or its history.

! Hijacking the shared demo login

Its email and password can't be changed, and it resets itself when idle.

Read across a row to follow one thread, or down a column to see one stage. Full detail with diagrams: **QuoteVoice-Product-Journey.pdf**

Demo login: **demo@quotevoice.test** · **QuoteVoice-demo-2026**